ALGEBRA

बीजगणित

SHEET-02

CLASS NOTES

CONCEPT OF PERFECT SQUARE

$* \quad (a-3)^2 = 0$

$\Rightarrow \quad (a-3) = 0$

$\Rightarrow \quad a = 3$

$* \quad (a+4)^2 = 0$

$\Rightarrow \quad (a+4) = 0$

$\Rightarrow \quad a = -4$

$* \quad (a-3)^2 + (b+4)^2 = 0 \qquad a=? \qquad b=?$

$+ \quad + \quad + \quad = +ve \times$

$0 \quad + \quad 0 \quad = 0$

Sol^n $a = 3 \qquad b = -4$

$* \quad (a-3)^2 + (b+1)^2 + (c-2)^2 = 0$

$0 \qquad 0 \qquad 0$

find $2a+3b-4c = ?$

Soln

$a = 3 \qquad b = -1 \qquad c = 2$

$2a+3b-4c = 6-3-8$

$= -5$

1. If $(a-3)^2 + (b-4)^2 + (c-9)^2 = 0$, then $\sqrt{a+b+c}$ = ?

(a) –4 (b) 4

(c) ±4 (d) ±2

$a = 3 \quad b = 4 \quad c = 9$

$\sqrt{a+b+c} = \sqrt{3+4+9} = \sqrt{16}$

$= 4$

$$x^2 = 16$$

$$\Rightarrow x = \pm\sqrt{16}$$

$$\Rightarrow \boxed{x = \pm 4}$$

$$x = \sqrt{16}$$

$$\therefore \boxed{x = 4}$$

2. If $(a-1)^2 + (b+2)^2 + (c+1)^2 = 0$, then find $2a - 3b + 7c = ?$

(a) 12 (b) −11

(c) 3 (d) 1

$a = 1 \quad b = -2 \quad c = -1$

$\therefore 2a - 3b + 7c$

$= 2 + 6 - 4$

$= 1$

3. If $(a-4)^2 + (b-5)^2 + (c-3)^2 = 0$, then the value of $\frac{a+b}{c}$ is:

(a) 0 (b) 3

(c) 1 (d) –3

$a = 4 \quad b = 5 \quad c = 3$

$\therefore \frac{a+b}{c} = \frac{4+5}{3} = 3$

4. If $(x + y - z)^2 + (y + z - x)^2 + (z + x - y)^2 = 0$, then $x + y + z = ?$

(a) $\sqrt{3}$ (b) $3\sqrt{3}$

(c) 3 (d) 0

$x+y-z = 0$

(+) $y+z-x = 0$

(+) $z+x-y = 0$

$x+y+z = 0$

Q. $(a+2)^2+(b-3)^2=0$

$a=?$ $b=?$

Ans

$a+2=0 \Rightarrow a=-2$

$b-3=0 \Rightarrow b=3$

$(a+2)^2+(b-3)^2=0$ ✓

$a^2+4+4a+b^2+9-6b=0$

$\Rightarrow a^2+b^2=-4a+6b-13$

Q. $a^2+b^2=2(-2a+3b)-13$

$a=?$ $b=?$

Ans

Condition

✓ i) Variables → right side

✓ ii) 2(Variable)

$a=\frac{-2}{1}$ $b=\frac{+3}{1}$

5. If $a^2 + b^2 + c^2 = 2(a - b - c) - 3$, then the value of $4a - 3b + 5c$ is –

(a) 2 (b) 3

(c) 5 (d) 6

$a = 1 \quad b = -1 \quad c = -1$

$4a - 3b + 5c$

$= 4 + 3 - 5$

$= 2$

i) V → rig

ii) Q(v)

6. If $a^2 + b^2 + 49c^2 + 18 = 2(b-28c-a)$ then the value of $(a+b-7c)$ is:

SSC CGL 12/04/2022 (Shift- 2)

(a) 4 (b) 3

(c) 2 (d) 1

$a = \frac{-1}{1}$ $b = \frac{1}{1}$ $c = -\frac{28}{49} = -4$

$\therefore a+b-4c$

$= -1+1-4\left(\frac{-4}{4}\right)$

$= 0+4$

V → right side

2(V)

7. If $a^2 + b^2 + 49c^2 + 18 = 2(b - 28c - a)$, then the value of $(a - b - 7c)$ is:

SSC CGL 18/04/2022 (Shift- 2)

(a) 2
(b) 1
(c) 3
(d) 4

$a = \frac{-1}{1}$ $b = \frac{1}{1}$ $c = -\frac{28}{49} \frac{4}{7}$

$a - b - 7c$

$= -1 - 1 - 7 \times \left(\frac{-4}{7}\right)$

$= -2 + 4$

$= 2$

V → right side

2(V)

8. If $a^2 + b^2 + 49c^2 + 18 = 2(b + 28c - a)$, then the value of $(2a - b + 7c)$ is:

SSC CGL 19/04/2022 (Shift- 2)

(a) – 3 (b) 1

(c) – 4 (d) 5

$a = \frac{-1}{1}$ $b = \frac{1}{1}$ $c = \frac{28}{49} \frac{4}{7}$

$2a - b + 7c$

$= -2 - 1 + 7 \times \frac{4}{7}$

$= -3 + 4$

$= 1$

9. If $a^2 + b^2 + c^2 = 2(a + c - 1)$, then the value of $a^3 + b^3 + c^3$ = ?

SSC CGL 12/09/2024 (Shift-03)

(a) 0 (b) 2
(c) 4 (d) 1

$a^2+b^2+c^2 = 2(a+c-1)$

$a = \frac{1}{1}$ $b = \frac{0}{1}$ $c = \frac{1}{1}$

V → right

$a^3+b^3+c^3$

$= 1+0+1$

$= 2$

QUESTIONS BASED ON $(A + B + C)^2$

* $(a+b)^2 = a^2+b^2+2(ab)$

* $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$

* $(a-b+c)^2 = a^2+b^2+c^2+2(-ab-bc+ac)$

* $(a+b-c)^2 = a^2+b^2+c^2+2(ab-bc-ac)$

* $(a-b-c)^2 = a^2+b^2+c^2+2(-ab+bc-ac)$

10. If $(x + y + z) = 23$ and $x^2 + y^2 + z^2 = 179$ then find the value of $(xy + yz + zx)$.

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 225 (b) 280

(c) 175 (d) 350

$(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$

$\Rightarrow 529 = 179 + 2 \times R$

$\Rightarrow 350 = 2R$

175

10. If $(x + y + z) = 23$ and $x^2 + y^2 + z^2 = 179$ then find the value of $(xy + yz + zx)$.

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 225 (b) 280

(c) 175 (d) 350

$x+y=23$ $\quad x^2+y^2=179$

$xy=?$

$\therefore (x+y)^2 = x^2+y^2+2xy$

$\Rightarrow 529 = 179 + 2xy$

$\Rightarrow 350 = 2xy$

$175 = xy$

11. If $a + b - c = 5$ and $ab - bc - ac = 10$, then find the value of $a^2 + b^2 + c^2$.

SSC CGL 18/04/2022 (Shift- 01)

(a) 15 (b) 45

(c) 5 (d) 40

$$(a+b-c)^2 = a^2+b^2+c^2 + 2(ab-bc-ac)$$

$$\Rightarrow 25 = k + 2\times 10$$

$$\Rightarrow S = k$$

$x^2 = 9$

$x = \pm\sqrt{9}$

$x = \pm 3$

12. If $a^2 + b^2 + c^2 = 6.25$ and $(ab+bc+ca) = 0.52$, what is the value of $(a+b+c)$, if $(a+b+c) < 0$?

SSC CGL 11/04/2022 (Shift- 03)

(a) ± 2.7 (b) -2.7

(c) -2.8 (d) ± 2.8

$(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$

$= 6.25 + 2 \times 0.52$

$= 6.25 + 1.04$

$(a+b+c)^2 = 7.29$

$\Rightarrow a+b+c = \pm 2.7$

Ans $a+b+c = -2.7$ $\because a+b+c < 0$

13. If $x + y + z = 13$, $x^2 + y^2 + z^2 = 91$ and $xz = y^2$, then the difference between z and x is:

SSC CHSL 05/08/2021 (Shift- 01)

(a) 3 (b) 8

(c) 5 (d) 9

$\therefore (x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$

$169 = 91 + 2(xy+yz+y^2)$

$\Rightarrow 39 = y(x+z+y)$

$\Rightarrow 3 \times 13 = y \times 13$

$y = 3$

$x+z = 10$ $\quad xz = 9$

$9 + 1$ $\quad 9 \times 1$

diff $|z-x| = |1-9| = 8$

13. If $x + y + z = 13$, $x^2 + y^2 + z^2 = 91$ and $xz = y^2$, then the difference between z and x is:

SSC CHSL 05/08/2021 (Shift- 01)

(a) 3 (b) 8

(c) 5 (d) 9

$x+y+z=13$ $1+3+9$

$x^2+y^2+z^2=91$ $1+9+81$

$xz=y^2$ $1\times9=(3)^2$

$1|8$

14. If $a + b + c = 3$, $a^2 + b^2 + c^2 = 6$ and $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$, Here, a, b & c are non-negative, then abc = ?

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$

(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

$\therefore (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$

$\Rightarrow 9 = 6 + 2abc$

$\Rightarrow 3 = 2abc$

$\Rightarrow \frac{3}{2} = abc$

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

$\Rightarrow \frac{bc + ac + ab}{abc} = 1$

$\Rightarrow ab + bc + ac = abc$

QUESTIONS BASED ON
$(a^3 + b^3 + c^3 - 3abc)$

$a^3+b^3+c^3-3abc$

F-1 → $(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$

F-2 → $(a+b+c)\left[(a+b+c)^2-3(ab+bc+ac)\right]$

F-3 → $\frac{(a+b+c)}{2}\left[3(a^2+b^2+c^2)-(a+b+c)^2\right]$

F-4 → $\frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$

a = _ _ _

b = _ _

c = _ _

15. If $a + b + c = 6$, $a^2 + b^2 + c^2 = 14$ and $ab + bc + ca = 11$, then what is the value of $a^3 + b^3 + c^3 - 3abc$?

SSC CGL 02/12/2022 (Shift-03)

(a) 31 (b) 12

(c) 18 (d) 42

$$a^3+b^3+c^3-3abc = (a+b+c)\left[a^2+b^2+c^2-ab-bc-ac\right]$$

$$= 6\left[14-11\right]$$

$$= 6 \times 3$$

$$= 18$$

16. If $a + b + c = 5$, $a^3 + b^3 + c^3 = 85$ and $abc = 25$, then find the value of $a^2 + b^2 + c^2 - ab - bc - ca$.

(a) 2 (b) 4

(c) 6 (d) 8

$a^3+b^3+c^3-3abc = (a+b+c)[a^2+b^2+c^2-ab-bc-ac]$

$\Rightarrow 85 - 75 = 5(k)$

$\Rightarrow 10 = 5k$

$2 = k$

17. **Factorize the given algebraic expression.**

$x^3 + 27y^3 + 64z^3 - 36xyz$

~~(a)~~ $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 + 3xy + 12yz + 4xz)$

~~(b)~~ $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 12xy^2 - 3yz - 4xz)$

~~(c)~~ $(x - 3y - 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$

(d) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$

$x^3+(3y)^3+(4z)^3-3\times x\times 3y\times 4z = (x+3y+4z)(x^2+9y^2+16z^2-3xy-12yz-4xz)$

18. What is the value of

$$\frac{\left[(2.247)^3 + (1.730)^3 + (1.023)^3 - 3 \times 2.247 \times 1.730 \times 1.023\right]}{\left[(2.247)^2 + (1.730)^2 + (1.023)^2 - 2.247 \times 1.730 - 1.730 \times 1.023 - 2.247 \times 1.023\right]}$$

(a) 1.730
(b) 4
(c) 5
(d) 5.247

$a = 2.247$, $b = 1.730$, $c = 1.023$

$$\frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2-ab-bc-ac} = \frac{(a+b+c)(a^2+b^2+c^2-ab-bc-ac)}{(a^2+b^2+c^2-ab-bc-ac)}$$

$= 5$

$2.247 + 1.730 + 1.023 = 5.000$

19. If $a + b + c = 15$ and $ab + bc + ca = 35$, then find the value of $a^3 + b^3 + c^3 - 3abc$.

SSC CPO 29/06/2024 (Shift-01)

(a) 2100 (b) 1500

(c) 1200 (d) 1800

$a^3+b^3+c^3-3abc = (a+b+c)\left[(a+b+c)^2 - 3(ab+bc+ac)\right]$

$= 15\left[225-105\right]$

$= 15 \times 120$

$= 1800$

20. If $(a + b + c) = 14$, and $(a^3 + b^3 + c^3 - 3abc) = 98$, find the value of $(ab + bc + ca)$.

SSC CGL 26/07/2023 (Shift-04)

(a) 60 (b) 64

(c) 65 (d) 63

$a^3+b^3+c^3-3abc = (a+b+c)\left[(a+b+c)^2-3(ab+bc+ca)\right]$

$\Rightarrow 98 = 14\,[196-3k]$

$\Rightarrow 3k = 189$

$k = 63$

21. If $(a + b + c) = 12$, and $(a^2 + b^2 + c^2) = 50$, find the value of $(a^3 + b^3 + c^3 - 3abc)$.

SSC CGL 19/07/2023 (Shift-02)

(a) 36 **(b) 24**

(c) 42 **(d) 48**

$$a^3+b^3+c^3-3abc = \frac{(a+b+c)}{2}\left[3(a^2+b^2+c^2)-(a+b+c)^2\right]$$

$$= 6\ [3\times 50 - 144]$$

$$= 6\times 6$$

$$= 36$$

21. If $(a + b + c) = 12$, and $(a^2 + b^2 + c^2) = 50$, find the value of $(a^3 + b^3 + c^3 - 3abc)$.

SSC CGL 19/07/2023 (Shift-02)

(a) 36
(b) 24
(c) 42
(d) 48

$a+b=12$ $\quad a^2+b^2=50$ $\quad a^3+b^3=?$

$(a+b)^2 = a^2+b^2+2ab$

$144 = 50+2ab$

$\Rightarrow 94 = 2ab$

47

$(a^3+b^3) = (a+b)\left[(a+b)^2-3ab\right]$

$= 12(144-141)$

$= 12 \times 3$

$= 36$

22. If $x + y + z = 2$, $x^3 + y^3 + z^3 - 3xyz = 74$, then $(x^2 + y^2 + z^2)$ is equal to :

SSC CGL 23/08/2021 (Shift- 03)

(a) 22 (b) 29

(c) 26 (d) 24

$$x^3+y^3+z^3-3xyz = \frac{(x+y+z)}{2}\left[3(x^2+y^2+z^2)-(x+y+z)^2\right]$$

$$\Rightarrow 74 = \frac{2}{2}\left[3\times k - 4\right]$$

$$\Rightarrow 78 = 3k$$

$$26 = k$$

23. If $x + y + z = 13$, $x^2 + y^2 + z^2 = 133$ and $x^3 + y^3 + z^3 = 847$, then the value of $\sqrt[3]{xyz}$ is:

SSC CPO 24/11/2020 (Shift- 02)

(a) 8 (b) 7

(c) −9 (d) −6

$$x^3+y^3+z^3-3xyz = \frac{(x+y+z)}{2}\left[3(x^2+y^2+z^2)-(x+y+z)^2\right]$$

$$\Rightarrow 847-3xyz = \frac{13}{2}\left[3\times133-169\right]$$

$$\Rightarrow 847-3xyz = \frac{13\times115}{1}$$

$$\Rightarrow 847-1495 = 3xyz$$

$$-648 = 3xyz$$

$$-216 = xyz$$

$$\therefore xyz = -216$$

$$\sqrt[3]{xyz} = -6$$

24. If $x = 110$, $y = 111$, $z = 112$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CPO 28/06/2024 (Shift-01)

(a) 999
(b) 991
(c) 997
(d) 995

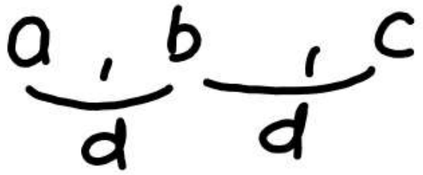
$$x^3+y^3+z^3-3xyz = \frac{(x+y+z)}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]$$

$$= \frac{333}{2}\left[(-1)^2+(-1)^2+(2)^2\right]$$

$$= \frac{333}{\cancel{2}} \times \cancel{6}\,3$$

$$= 999$$

* When a, b, c are in A.P



$$a^3+b^3+c^3-3abc = \frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$$= 3b \times 3d^2$$

$$= 9bd^2$$

24. If $x = 110$, $y = 111$, $z = 112$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CPO 28/06/2024 (Shift-01)

(a) 999
(b) 991
(c) 997
(d) 995

$x^3+y^3+z^3-3xyz = 9bd^2$

$= 9 \times 111 \times 1^2$

$= 999$

25. If $x = 222$, $y = 223$ and $z = 224$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

SSC CGL 12/12/2022 (Shift-04)

(a) 2007 (b) 2004

(c) 2006 (d) 2005

ans = $9bd^2$

$= 9 \times 223 \times 1$

$= 2007$

26. Find $a^3 + b^3 + c^3 - 3abc$ if $a = 1001$, $b = 1004$ and $c = 1007$.

(a) 80235 (b) 81234

(c) 79356 (d) 81324

$a = 1001 \quad b = 1004 \quad c = 1007$

$3 \qquad 3$

$a^3+b^3+c^3-3abc = 9bd^2$

$= 9 \times 1004 \times 9$

$= 1004 \times 81$

$= 81324$

27. If $x = z = 225$, and $y = 226$ then $x^3 + y^3 + z^3 - 3xyz = ?$

(a) 765 (b) 676

(c) 576 (d) 674

$x=225 \quad z=225 \quad y=226$

$0 \qquad 1$

$$x^3+y^3+z^3-3xyz = \frac{(x+y+z)}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]$$

$$= \frac{676}{2}\left[(-1)^2+1^2+0\right]$$

$$= \frac{676}{2} \times 2$$

$$a^3+b^3+c^3-3abc$$

$$(a+b+c)(a^2+b^2+c^2-ab-bc-ac) = \frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$$\therefore\ (a^2+b^2+c^2-ab-bc-ac) = \frac{1}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$$= 3d^2 \quad (A.P)$$

28. If a = 2022, b = 2021 and c = 2020, then value of $a^2 + b^2 + c^2 - ab - bc - ca$ is:

SSC CHSL 26/05/2022 (Shift- 03)

(a) 2
(b) 4
(c) 3
(d) 1

$= \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$

$= 3d^2$

$= 3 \times (-1)^2$

$= 3$

29. If $p = 106$, $q = 105$ and $r = 104$, then value of $p^2 + q^2 + r^2 - pq - qr - rp$ is:

SSC Phase-XII 25/06/2024 (Shift-03)

(a) 2
(b) 0
(c) 1
(d) 3

$= 3d^2$
$= 3 \times (-1)^2$
$= 3$

30. If a = 101, b = 102 and c = 103, then $a^2 + b^2 + c^2 - ab - bc - ca$

SSC CGL 24/07/2023 (Shift-01)

(a) 2
(b) 4
(c) 3
(d) 6

ans = $3d^2$
$= 3 \times 1$
$= 3$

31. If $x = 32$, $y = 33$ and $z = 35$, then evaluate the expression $x^3 + y^3 + z^3 - 3xyz$.

H.W Try

SSC CHSL 01/06/2022 (Shift- 03)

(a) 1120
(b) 1000
(c) 900
(d) 700

F-4

32. If $ab + bc + ca = 8$ and $a^2 + b^2 + c^2 = 20$, then possible value of $\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$ is :

(a) 72 (b) 56

(c) 84 (d) 80

$(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$

$= 20 + 2\times 8$

$= 36$

$\therefore (a+b+c) = 6$

$$(a+b+c)\left[a^2+b^2+c^2-(ab+bc+ac)\right] = \frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$\Rightarrow 6[20-8] = \,''$

$\Rightarrow 6\times 12 =$

$\Rightarrow 72 = \,''$

$= \,''$

33. If $(a + b + c) = 19$, and $(a^2 + b^2 + c^2) = 155$, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.

SSC CGL 18/07/2023 (Shift-02)

(a) 104 (b) 108

(c) 100 (d) 98

$$\frac{(a+b+c)}{2}\left[3(a^2+b^2+c^2)-(a+b+c)^2\right] = \frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$= 3 \times 155 - 19^2$

$= 465 - 361$

$= 104$

34. If $(a^3 + b^3 + c^3 - 3abc) = 405$, and $(a + b + c) = 15$, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.

SSC CGL 20/07/2023 (Shift-01)

(a) 27 (b) 54

(c) 18 (d) 45

$$a^3+b^3+c^3-3abc = \frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

$$\Rightarrow 405 = \frac{15}{2} \times R$$

$S4 = R$

35. If $(a^3 + b^3 + c^3 - 3abc) = 405$, and $(a - b)^2 + (b - c)^2 + (c - a)^2 = 54$, find the value of $(a + b + c)$.

Try.
H.W

SSC CGL 26/07/2023 (Shift-02)

(a) 15
(b) 45
(c) 9
(d) 27

36. Simplify the following expression.

$$\frac{(59\times59\times59)+(54\times54\times54)+(57\times57\times57)-3(59)(54)(57)}{(59+54+57)}$$

SSC CHSL 27/05/2022 (Shift- 01)

(a) 38 (b) 76

(c) 170 (d) 19

$$\frac{a^3+b^3+c^3-3abc}{(a+b+c)} = \frac{\frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]}{(a+b+c)}$$

$$= \frac{1}{2}\left[2S+9+4\right] = \frac{38}{2} = 19$$

37. If $a = 25, b = 15, c = -10$, then

$$\frac{a^3+b^3+c^3-3abc}{(a-b)^2+(b-c)^2+(c-a)^2} = ?$$

(a) 30 (b) −15

(c) −30 (d) 15

$$= \frac{\frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]}{\left[(a-b)^2+(b-c)^2+(c-a)^2\right]}$$

$$= \frac{30}{2} \text{ IS}$$

$$* \underbrace{a^3+b^3+c^3-3abc} \overset{0}{=} \underbrace{\frac{(a+b+c)}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]}$$

$$\overset{0}{\left(\frac{a+b+c}{2}\right)} \overset{0}{\left[(a-b)^2+(b-c)^2+(c-a)^2\right]} = 0$$

$$\frac{a+b+c}{2} = 0$$

$$\therefore \boxed{a+b+c=0}$$

$$(a-b)^2+(b-c)^2+(c-a)^2 = 0$$

$$a-b=0 \Rightarrow \boxed{a=b}$$
$$b-c=0 \Rightarrow \boxed{b=c}$$
$$c-a=0 \Rightarrow \boxed{c=a}$$

$$\Rightarrow a=b=c$$

If $a^3+b^3+c^3-3abc=0$ or $a^3+b^3+c^3=3abc$ then i) $a+b+c=0$ ii) $a=b=c$

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* $a^3+b^3+c^3-3abc=0$ $\quad a+b+c=0$
* $a^3-b^3+c^3+3abc=0$ $\quad a-b+c=0$
* $a^3+b^3-c^3+3abc=0$ $\quad a+b-c=0$

38. If $a^3 - b^3 - c^3 - 3abc = 0$, then

(a) $a = b = c$ (b) $a + b + c = 0$

(c) $a + c = b$ (d) $a = b + c$

$a^3+b^3+c^3-3abc=0$

$a^3-b^3-c^3-3abc=0$

$a+b+c=0$

$a-b-c=0$

$\Rightarrow a=b+c$

39. If $a + 2b + 3c = 0$, then what is the value of $(a^3 + 8b^3 + 27c^3)$?

SSC CPO 28/06/2024 (Shift-01)

(a) 18 abc
(b) 12 abc
(c) 0
(d) 6 abc

$a+b+c=0$

$a+2b+3c=0$

$a^3+b^3+c^3=3abc$

$a^3+8b^3+27c^3=3\times a\times 2b\times 3c$

$=18abc$

40. If $a + b = c$, then the expression $a^3 + b^3 - c^3 + 3abc$ will be equal to:

SSC CGL 24/09/2024 (Shift-03)

(a) 0
(b) 3
(c) 2
(d) 1

$a+b+c=0$

$a+b-c=0$

$a^3+b^3+c^3-3abc=0$

$a^3+b^3-c^3+3abc=0$

41. If $8a^3 + b^3 + 27c^3 = 18\,abc$, then the relation among a, b and c is:

SSC CGL 17/09/2024 (Shift-02)

(a) $b + 3c = 2a$ (b) $2a + b = 3c$

(c) $2a + 3c = b$ (d) $b + 3c = -2a$ ✓

$a^3+b^3+c^3 = 3abc$

$a+b+c=0$

$(2a)^3+(b)^3+(3c)^3 = 3\times 2a\times b\times 3c$

$= 18abc$

$2a+b+3c=0$

$\Rightarrow b+3c=-2a$

* $(a)^3+(b)^3+(c)^3 = 3abc$ $\quad a+b+c=0$

* $(x-y)^3+(y-z)^3+(z-x)^3 = 3(x-y)(y-z)(z-x)$

$\therefore \cancel{x-y}+\cancel{y-z}+\cancel{z-x} = 0$

* $(2a-2b)^3+(2b-2c)^3+(2c-2a)^3 = 3\times(2a-2b)(2b-2c)(2c-2a)$

42. Simplify the following expression.

$$\frac{(x-y)^3+(y-z)^3+(z-x)^3}{(x-y)(y-z)(z-x)}$$

SSC CGL 13/09/2024 (Shift-01)

(a) 3 (b) 1

(c) 0 (d) 2

$$\frac{3(x-y)(y-z)(z-x)}{(x-y)(y-z)(z-x)}$$

43. Find the value of $(1.6)^3 - (0.9)^3 - (0.7)^3$.

(a) 3.24 (b) –3.24

(c) 3.024 (d) –3.024

$(1.6)^3 + (-0.9)^3 + (-0.7)^3 = 3(1.6)(-0.9)(-0.7)$

$1.6 - 0.9 - 0.7 = 0$

$= +\ 3.024$

44. If $(3x + 1)^3 + (x - 3)^3 + (4 - 2x)^3 + 6(3x + 1)(x - 3)(x - 2) = 0$, then what is the value of x?

(a) -1 (b) $-\frac{1}{2}$

(c) 1 (d) $\frac{1}{2}$

$(3x+1)^3+(x-3)^3+(4-2x)^3-3(3x+1)(x-3)(4-2x)=0$ ($-2(x-2)$)

$=(3x+1)^3+(x-3)^3+(4-2x)^3+6(3x+1)(x-3)(x-2)=0$

$a^3+b^3+c^3-3abc=0$

i) $a+b+c=0$

ii) $a=b=c$

44. If $(3x + 1)^3 + (x - 3)^3 + (4 - 2x)^3 + 6(3x + 1)(x - 3)(x - 2) = 0$, then what is the value of x?

(a) -1 (b) $-\frac{1}{2}$

(c) 1 (d) $\frac{1}{2}$

$(3x+1)^3+(x-3)^3+(4-2x)^3 - - - - - = 0$

then

$3x+1+x-3+4-2x=0$

$\Rightarrow 2x+2=0$

$\Rightarrow 2x=-2$

$\therefore x=-1$

$(4x-5)^3+(x-2)^3+(6x-15)^3=$

$4x-5+x-2+6x-15=0$

$\Rightarrow 11x=22$

$x=2$

$\therefore 2+\frac{3}{2}=\frac{7}{2}$

45. If $(4x-5)^3+(x-2)^3+27(2x-5)^3=9(4x-5)(x-2)(2x-5)$, then the value of $\left(x+\frac{3}{2}\right)$ will be:

SSC CHSL 05/08/2021 (Shift- 02)

(a) $\frac{1}{2}$

(b) $\frac{5}{2}$

(c) $\frac{3}{2}$

(d) $\frac{7}{2}$

$(2x)^3=8(x)^3$

$(3a)^3=27(a^3)$

46. If $(5x + 1)^3 + (x - 3)^3 + 8(3x - 4)^3 = 6(5x + 1)(x - 3)(3x - 4)$, then x is equal to:

(a) $\frac{5}{6}$ (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

$(5x+1)^3+(x-3)^3+(6x-8)^3 = - - - -$

$5x+1+x-3+6x-8=0$

$\Rightarrow 12x = 10$

$x = \frac{5}{6}$

QUESTIONS BASED ON $(a^3 + b^3)$ and $(a^3 - b^3)$

* a^3+b^3

$(a+b)(a^2+b^2-ab)$ ✓

or

$(a+b)[(a+b)^2-3ab]$ ✓

* a^3-b^3

$(a-b)(a^2+b^2+ab)$ ✓

or

$(a-b)[(a-b)^2+3ab]$

Q $a^2+b^2-ab=7$ $a+b=8$ $a^3+b^3=?$

Ans $(a^3+b^3) = (a+b)(a^2+b^2-ab)$

$= 8 \times 7$

$= 56$

Q. $a^2+b^2-ab = 10$ $a+b=9$ $a^3+b^3=?$

Ans $a^3+b^3 = (a+b)(a^2+b^2-ab)$

$= 9 \times 10$

$= 90$

Q. $a^2+b^2+ab=7$ $\quad a-b=5$ $\quad a^3-b^3=?$

Ans $a^3-b^3=(a-b)(a^2+b^2+ab)$

$=5\times7$

$=35$

Q. $a^2+b^2+ab=?$ $\quad a-b=5$ $\quad a^3-b^3=10$

Ans

$a^3-b^3=(a-b)(a^2+b^2+ab)$

$2\ \cancel{10}=\cancel{5}\times k$

$2=k$

Q. $a+b=7$ $ab=11$ $a^3+b^3=?$

Ans. $a^3+b^3=(a+b)\left[(a+b)^2-3ab\right]$

$= 7\left[49-33\right]$

$= 7\times16$

$= 112$

Q. $a+b=8$ $ab=20$ $a^3+b^3=?$

Ans

$a^3+b^3=(a+b)\left[(a+b)^2-3ab\right]$

$= 8\left[64-60\right]$

$= 8\times4 = 32$

Q. $a-b=10$ $ab=20$ $a^3-b^3=?$

Ans $a^3-b^3=(a-b)\left[(a-b)^2+3ab\right]$

$= 10\ [100+60]$

$= \boxed{1600}$ ✓

Q. $a-b=20$ $ab=5$ $a^3-b^3=?$

Ans $a^3-b^3=(a-b)\left[(a-b)^2+3ab\right]$

$= 20\ [400+15]$

$= 20 \times 415$

$= \underline{8300}$

$a^3-b^3=(a-b)(a^2+b^2+ab)$

$36 \quad 216 = 6 \times k$

$k=36$

47. If $a^3 - b^3 = 216$ and $a - b = 6$, then $(a + b)^2 - ab$ is equal to :

(a) 38 **(b) 42**

(c) 52 **(d) 36**

$a^2+b^2+2ab-ab$

$= a^2+b^2+ab$

48. If $a - b = 5$ and $ab = 25$, then the value of $a^3 - b^3$ is:

SSC Phase-XII 25/06/2024 (Shift-02)

(a) 525 (b) 500

(c) 450 (d) 400

$$a^3 - b^3 = (a-b)\left[(a-b)^2 + 3ab\right]$$

$$= 5\left[25 + 75\right]$$

$$= 500$$

49. If $(a - b) = 9$, and $(a^3 - b^3) = 4410$, find the value of ab.

SSC CHSL 02/08/2023 (Shift-02)

(a) 190 (b) 112

(c) 162 (d) 136

$a^3 - b^3 = (a-b)[(a-b)^2 + 3ab]$

$4410 = 9[81 + 3ab]$

490

$\Rightarrow 409 = 3ab$

$ab = 136.33$

50. If $a + b = 10$ and $ab = 6$, then the value of $a^3 + b^3$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 860 (b) 820

(c) 800 (d) 840

$$a^3+b^3 = (a+b)\left[(a+b)^2-3ab\right]$$

$$= 10\left[100-18\right]$$

$$= 820$$

* $(a+b)=10 \quad a^3+b^3=20 \quad ab=?$

Ans

$(a^3+b^3)=(a+b)\left[(a+b)^2-3ab\right]$

$\Rightarrow 2\not{0} = 1\not{0}\,[100-3ab]$

$\Rightarrow 3ab=98$

$ab=\frac{98}{3}$

Q. $2x+3y=10 \quad 8x^3+27y^3=30$

$xy=?$

Ans $(8x^3+27y^3)=(2x+3y)\left[(2x+3y)^2-3X6\right.$

$\Rightarrow 3\not{0}=1\not{0}\,[100-18xy]$

$\Rightarrow 18xy=97$

$xy=\frac{97}{18}$

51. If $3a + 2b = 27$ and $27a^3 + 8b^3 = 1458$, then find $2ab$.

SSC CGL 18/09/2024 (Shift-03)

(a) 72 **(b) 70**

(c) 77 **(d) 75**

$(3a)^3 + (2b)^3 = (3a+2b)\left[(3a+2b)^2 - 3\times 3a\times 2b\right]$

$\Rightarrow 1458 = 27\left[729 - 18ab\right]$

$\Rightarrow 18ab = 648$

$\div 9$ $\quad 2ab = 72$ $\quad \div 9$

$x^3+y^3=(x+y)(x^2+y^2-xy)$

$= 2 \times 0$

$= 0$

52. If $\frac{x}{y}+\frac{y}{x} = 1$ and $x + y = 2$, then the value of $x^3 + y^3$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 0 (b) 1

(c) 3 (d) 2

$\frac{x}{y}+\frac{y}{x} = 1$

$\Rightarrow \frac{x^2+y^2}{xy} = 1$

$\Rightarrow x^2+y^2 = xy$

$\Rightarrow x^2+y^2-xy = 0$

$$* (a^3 - b^3) = (a-b)(a^2 + b^2 + ab)$$

$$\Rightarrow (a^3 - b^3) \div (a-b) = (a^2 + b^2 + ab)$$

53. If $(8x^3 - 27y^3) \div (2x - 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of $(2A + B - C)$ is :

(a) 4 (b) 6

(c) 5 (d) 3

$[(2x)^3 - (3y)^3] \div (2x - 3y) = 4x^2 + 6xy + 9y^2$

$Ax^2 + Bxy + Cy^2$

$(a^3 - b^3) \div (a - b) = (a^2 + ab + b^2)$

$A = 4 \quad B = 6 \quad C = 9$

$\therefore 2A + B - C$

$= 8 + 6 - 9$

$= 5$

53. If $(8x^3 - 27y^3) \div (2x - 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of $(2A + B - C)$ is :

(a) 4 (b) 6

(c) 5 (d) 3

4 +6 9

A=4 B=6 C=9

2A+B-C

= 8+6-9

= 5

54. If $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = \left(\sqrt{2}x - \sqrt{3}y\right)(Ax^2 + By^2 + Cxy)$, then the value of $A^2 + B^2 - C^2$ is :

$2x^2 + 3y^2 + \sqrt{6}xy$

$4 + 9 - 6 = 7$

(a) 11 (b) 7

(c) 19 (d) 10

$4x^2 - 6xy \quad 9y^2$

55. If $(8x^3 + 27y^3) \div (2x + 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of $(5A + 4B + 3C)$ is :

$20 - 24 + 27$

(a) 26 (b) 23

(c) 24 (d) 27

$A=4 \quad B=-6 \quad C=9$

$(\sqrt{5}x)^3 + (\sqrt{2}y)^3$

56. If $5\sqrt{5}x^3 + 2\sqrt{2}y^3 = (Ax + \sqrt{2}y)(Bx^2 + 2y^2 + Cxy)$, then the value of $(A^2 + B^2 - C^2)$ is

$\sqrt{5}x + \sqrt{2}y \quad 5x^2 \quad 2y^2 - \sqrt{10}xy$

$5 + 25 - 10$

(a) 15 (b) 20

(c) 30 (d) 40

$A = \sqrt{5} \quad B = 5 \quad C = -\sqrt{10}$

$(3x+3y)^3 - (2x-2y)^3 = [(3x+3y)-(2x-2y)][(3x+3y)^2+(2x-2y)^2 + (3x+3y)(2x-2y)]$

$= (x+5y)$

57. ***

If $27(x+y)^3 - 8(x-y)^3 = (x+5y)(Ax^2 + By^2 + Cxy)$, then what is the value of $(A + B - C)$?

CGL 2019 Tier II (15/11/2020)

(a) 13 (b) 16

(c) 18 (d) 11

$a^3 - b^3 = (a-b)(a^2+b^2+ab)$

$(3x+3y)^3 - (2x-2y)^3$

a b

$[(3x+3y)^2 + (2x-2y)^2 + (3x+3y)(2x-2y)$

57. ***

If $27(x+y)^3-8(x-y)^3=(x+5y)(Ax^2+By^2+Cxy)$, then what is the value of (A + B – C)?

CGL 2019 Tier II (15/11/2020)

(a) 13 **(b) 16**

(c) 18 **(d) 11**

$x^2 \rightarrow 9+4+6 = 19 \leftarrow A$

$y^2 \rightarrow 9+4-6 = 7 \leftarrow B$

$xy \rightarrow 18-8 = 10 \leftarrow C$

$19+7-10$

$= 16$

$(3x+3y)^3 - (2x-2y)^3$

57. If $27(x+y)^3-8(x-y)^3=(x+5y)(Ax^2+By^2+Cxy)$, then what is the value of $(A+B-C)$?

19 7 10

$19+7-10$

CGL 2019 Tier II (15/11/2020)

(a) 13 (b) 16

(c) 18 (d) 11

$x^2 \rightarrow 9+4+6 = 19$

$y^2 \rightarrow 9+4-6 = 7$

$xy \rightarrow 18-8 = 10$